

# PHANIDHAR AKULA

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## SKILLS

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- TypeScript, React, Redux, Next.js, JavaScript, Node.js, Python, LLMs, RAG, REST APIs, PostgreSQL
- Supabase, MySQL, SQL, Tailwind CSS, FastAPI, Prompt Engineering, Vector Search, OAuth, Vercel, AWS
- Capacitor, iOS, Android, Git, GitHub Actions, CI/CD, Docker, pytest, Java, C++, Linux, OpenMP, Agile

## WORK EXPERIENCE

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**Full-Stack Engineer:** *LumiAI (Independent Product) – [studywithlumi.com](https://studywithlumi.com)* 10/2025 – Present

- Built and shipped a production React/TypeScript AI tutoring platform used by 220+ active students, generating document-grounded chat, quizzes, flashcards, spaced repetition, and voice tutoring.
- Engineered a serverless LLM inference layer on Vercel with token-verified requests, response streaming, and prompt caching, blocking 100% of anonymous inference calls and keeping provider keys server-side.
- Architected the full-stack data layer on PostgreSQL with row-level security across every user table, Google OAuth, protected server functions, and signed storage so users only read their own documents.
- Extended the platform to iOS and Android using Capacitor, wrapping the existing TypeScript codebase in native shells and submitting both builds for store review from a single shared repository.

**Computer Science Researcher:** *Miami University* 08/2024 – 07/2026

- Designed a Python-driven 4-city, 16-configuration HPC study on the OSC Pitzer cluster, cutting LA synthetic-home misassignment from 69.9% to 5.5% and lifting worst-case  $R^2$  from 0.812 to 0.917.
- Diagnosed and fixed OSM building-classification defects in Cityscape's open-source C++ pipeline across 5 metros, shipping 3 merged PRs that removed 1.25M stranded-population artifacts.
- Developed SimForge as M.S. thesis, an open-source Python framework unifying 3 traffic simulators (SUMO, MATSim, DTALite) behind one canonical data pipeline with containerized, deterministic execution.
- Accelerated shared-route pre-routing from 141.9 hours to 7.1 hours through deterministic route caching and 16-way parallelism, a 20 times speedup, then benchmarked 3 cities at up to 500K trips per seeded run.

**Software Engineer Intern:** *Smart Pathfinders Overseas Education* 01/2024 – 05/2024

- Collaborated with a team of 3 developers to build a private client portal in React and Redux, giving 180+ enrolled students one place to track applications, upload documents, and reach counselors.
- Implemented Python/MySQL REST APIs and relational data models behind role-based access control, so each student saw only their own records while staff managed the full caseload from one dashboard.

**Software Developer:** *Freelance – E-Commerce & Fitness Startup Clients* 05/2023 – 12/2023

- Delivered production full-stack software, building React and Redux interfaces against Python/MySQL REST APIs and deploying to AWS, replacing manual spreadsheet workflows for 300+ daily users.
- Migrated 10k+ records from client Excel workbooks into MySQL with a Python ETL script, validating row counts and key fields against the source so both clients cut over in one pass with no manual re-entry.

## EDUCATION

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**Miami University** | *M.S. Computer Science, GPA: 3.86/4.00* 08/2024 – 08/2026

Relevant Coursework: Generative AI, Advanced Database Systems, Machine Learning, Software Testing

Publications: 2 first-author manuscripts under review at IEEE T-ITS and Frontiers (2026)

Leadership: President, Graduate Students of Color Association (110+ members) | Elected Graduate Student Representative, University Graduate Council | Program Lead, NAE Grand Challenges Scholars Program

**Malla Reddy University** | *B.Tech Computer Science* 08/2020 – 05/2024

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Artificial Intelligence, Full-Stack Web Development, Database Management Systems, Computer Networks, Operating Systems